Shantanu

ECE Undergraduate | Embedded Systems Engineer (ESP32, Sensors, IoT)
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Linkedin | **Portfolio** | **GitHub**

Education

- **Thapar Institute of Engineering and Technology** — Patiala, Punjab
  Bachelor of Technology - Electronics and Communication Engineering — **August 2025 - Present**
- **Thapar Institute of Engineering and Technology** — Patiala, Punjab
  Minor Degree - Computer Science and Engineering — **August 2026 - December 2027**

Skills Summary

- **Embedded Systems**: ESP32, Arduino, I2C, UART, SPI, MQTT, Real-time Processing
- **Programming**: C, C++ (Arduino), Python, JavaScript
- **Hardware / Electronics**: Sensors, Electromagnets, Power Electronics, GPS Modules, OLED Displays
- **Tools / Platforms**: Git, GitHub, KiCad, Arduino IDE, ESP-IDF, Vercel

Selected Projects & Open-Source Contribution

- **Adafruit CircuitPython - Wi-Fi API** — **C, CircuitPython**
  Open-source C contribution: Implemented Raspberry Pi `wifi.radio.ap_info` in my initial PR #11625, exposing SSID, BSSID, RSSI, channel, and auth mode; my approach was credited in merged PR #11646.
- **VIGIL-FUSE** — **Python, IoT Security**
  Built a Python CLI for embedded/IoT security triage; 7 native rules flag credential-like strings, TLS bypasses, and plaintext HTTP/MQTT indicators, producing evidence-backed JSON/SARIF reports.
- **Zephyr Station** — **ESP32, C++**
  Built an ESP32 room monitor with 7 sensor modules for temperature, humidity, pressure, and air quality; added OLED readings, RTC-stamped SD logs, threshold alerts, and a live dashboard.
- **Energy Consumption Analysis** — **Python, PCA, K-Means**
  Analyzed 200 synthetic household profiles across 365 days using 51 behavioral features; PCA retained 95.05% variance in 10 components, and K-Means recovered 4 archetypes (ARI 0.813, NMI 0.828).

Experience

- **Ragastra MV Lev** — Patiala, Punjab
  Hardware & Controls Engineer — **September 2025 - Present**
  - **Electromagnetic System Design**: Designed and tuned custom electromagnets for magnetic levitation prototype with optimized coil parameters.
  - **Power Electronics**: Designed and debugged SMPS-based power circuitry for inductive electromagnetic loads.
  - **Motor Control Systems**: Implemented Arduino-based synchronized dual stepper motor control with real-time actuation.
- **Grosity** — Patiala, Punjab
  Early-Stage Founding Engineer — **October 2025 - February 2026**
  - **Web Development**: Developed and deployed company website using AI-assisted development workflows (Claude, GitHub Copilot, Kiro IDE).
  - **Autonomous Systems R&D**: Contributed to embedded electronics and control logic for autonomous payload-delivery drone systems.
  - **Product Strategy**: Aligned engineering decisions with B2B and D2C product models and business strategy.

Ongoing Research

- **GHFCVs - Reverse Engineering Study** — **December 2025 - Present**
  Literature review and comparative analysis of DC-DC converter topologies with hands-on teardown of commercial GHFCV systems to understand power electronics architecture and operational principles.

Certifications & Achievements

- **Internet of Things** - MIT Professional Education x Santander Open Academy
- **Google: Artificial Intelligence and Productivity** - Google x Santander Open Academy
- **National Finalist** - E Summit Ideastorm 2026, IIT Roorkee